

SIMATS ENGINEERING
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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Department:	AIML	Date:	

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I G. Naveen Kumar Reddy (192525288) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Naveen Kumar Reddy

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS

COMPARATIVE ANALYSIS OF HIERARCHICAL MEMORY SYSTEMS

Q1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

1. Overview / Scope

Modern processors use a multi-level cache hierarchy (L1, L2, L3) positioned between the CPU core and main memory (DRAM) to bridge the large speed gap between the two. Each level trades off capacity, speed, and proximity to the core differently, and this comparison examines the three levels across latency, bandwidth, throughput, and their combined effect on overall performance.

2. Comparative Table

Parameter	L1 Cache	L2 Cache	L3 Cache
Location	Inside each core (private)	Per-core or per-cluster (private)	Shared across all cores (on-die)
Typical Size	32–64 KB (per core, split I/D)	256 KB – 1 MB (per core)	8 MB – 64 MB (shared)
Access Latency	~1–4 CPU cycles (~1 ns)	~10–12 cycles (~3–5 ns)	~30–40 cycles (~10–20 ns)
Bandwidth	Highest (several hundred GB/s per core)	High, lower than L1 (100–200 GB/s)	Moderate, shared among cores (50–100 GB/s aggregate)
Throughput	Very high per-core instruction/data supply	High, slightly reduced due to larger array	Lower per-access, but large capacity sustains bulk transfers
Technology	SRAM, fastest and most expensive per bit	SRAM, slightly denser/slower than L1	SRAM (sometimes eDRAM), largest and slowest of the three
Hit Rate Contribution	Filters most repeated/loop accesses	Captures medium working-set data missed by L1	Captures large working sets shared across cores, reduces DRAM traffic

3. Analytical Discussion

L1 cache is split into separate instruction and data caches placed directly next to the execution units, so it is optimized purely for the lowest possible latency; its small size keeps access time to just a few cycles, which is essential because the pipeline stalls whenever L1 misses. L2 cache is larger and unified (instructions and data together), accepting a higher latency in exchange for holding a bigger working set, which reduces the frequency of costly L3 or DRAM accesses. L3 cache is shared across all cores of the chip, has the largest capacity, and acts as a last line of defense before main memory; its higher latency is acceptable because it is only reached after both L1 and L2 miss, and its shared nature also enables efficient inter-core data sharing and cache coherence.

From a bandwidth perspective, bandwidth decreases as the level number increases because each level is physically farther from the core and must serve more consumers (L3 is shared by multiple cores simultaneously), while throughput follows a similar trend but is also influenced by hit rate: a well-sized L2 or L3 can sustain high effective throughput even at higher per-access latency because it prevents expensive round trips to DRAM.

4. Impact on Processor Performance

- The Average Memory Access Time (AMAT) is largely determined by L1 hit rate; even a 1–2% improvement in L1 hit rate can yield a measurable IPC (instructions per cycle) gain because L1 misses are on the critical path of every instruction.
- L2 acts as a latency-bandwidth buffer: it absorbs the majority of L1 misses so that only a small fraction of accesses ever reach L3 or DRAM, which is essential for maintaining pipeline throughput in out-of-order cores.

- L3 primarily affects multi-core scalability and worst-case latency; a larger, well-managed L3 reduces DRAM bandwidth pressure and inter-core coherence traffic, which is critical in multi-threaded and server workloads.
- Overall system performance is governed by the AMAT formula: $AMAT = HitTime(L1) + MissRate(L1) \times [HitTime(L2) + MissRate(L2) \times (HitTime(L3) + MissRate(L3) \times MemoryAccessTime)]$, showing that each level's latency and hit rate compound multiplicatively down the hierarchy.

5. Conclusion

L1, L2, and L3 form a deliberately graded trade-off between speed and capacity: L1 optimizes for minimum latency at the cost of size, L3 optimizes for capacity and sharing at the cost of latency, and L2 occupies the balanced middle ground. Their combined, hierarchical operation is what allows modern processors to approach the effective performance of a large, fast memory while only paying the cost of a small, expensive one.

Q2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

1. Overview / Scope

SRAM (Static RAM) and DRAM (Dynamic RAM) are the two dominant volatile memory technologies used in the memory hierarchy. SRAM is used almost exclusively for on-chip caches (L1–L3), while DRAM is used for main memory. Their differing cell designs directly drive their latency, bandwidth, throughput, cost, and density trade-offs.

2. Comparative Table

Parameter	SRAM	DRAM
Cell Structure	6-transistor (6T) flip-flop, no capacitor	1 transistor + 1 capacitor (1T-1C)
Refresh Requirement	Not required (stable while powered)	Requires periodic refresh (~64 ms cycle) due to capacitor leakage
Latency	Very low (~1–2 ns, single-digit cycles)	Higher (~50–100 ns, tens of cycles) due to row activation/precharge
Bandwidth	Very high, limited mainly by on-chip wiring	High aggregate bandwidth (e.g., DDR5 ~40–50 GB/s per channel) but per-access lower than SRAM
Throughput	High sustained throughput for small, frequent accesses	High burst throughput for large sequential transfers, lower for random small accesses
Density / Cost per bit	Low density, high cost per bit (larger cell)	High density, low cost per bit (smaller cell)
Power Consumption	Higher static leakage power, no refresh power	Lower per-cell static power, but refresh adds dynamic power overhead
Typical Use	L1/L2/L3 caches, register files	Main memory (DIMMs), frame buffers

3. Analytical Discussion

SRAM's six-transistor cell holds its state as long as power is supplied, eliminating the refresh overhead that DRAM requires; this makes SRAM access deterministic and very fast, which is why it is chosen for caches where every nanosecond of latency directly affects instruction throughput. DRAM's single-transistor-single-capacitor cell is far denser and cheaper per bit, but the capacitor's charge leaks over time, forcing a refresh operation roughly every 64 milliseconds; this refresh, combined with the multi-step row activation and column access protocol (RAS/CAS), is the main source of DRAM's higher latency.

In terms of bandwidth, SRAM's bandwidth is essentially bound only by on-chip interconnect width because it can be accessed randomly with uniform latency, whereas DRAM achieves high peak bandwidth mainly through burst-mode sequential access and wide buses (as in DDR/GDDR/HBM), but suffers bandwidth degradation on random, small, or bank-conflicting accesses.

4. Role in Cache vs. Main Memory Design

- Cache design exploits SRAM's low, uniform latency to keep the CPU pipeline fed without stalls, accepting its high cost and low density because caches are intentionally small.
- Main memory design exploits DRAM's high density and low cost per bit to provide gigabytes of capacity, accepting higher latency because main memory is accessed far less frequently than cache when the hierarchy is working well.
- Throughput in the overall system depends on this complementary pairing: SRAM caches absorb most accesses (keeping effective latency low), while DRAM supplies the bulk capacity needed to hold the full working set and back the caches on misses.

5. Conclusion

SRAM and DRAM are chosen for fundamentally different design goals — SRAM for speed at small scale (cache), DRAM for capacity at low cost (main memory). Their latency and bandwidth differences are a direct consequence of cell architecture (6T static vs. 1T-1C dynamic), and a well-designed memory hierarchy leverages both technologies together rather than treating either as universally superior.

Q3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

1. Overview / Scope

Virtual memory (implemented via paging) and cache memory are both hierarchical management techniques, but they operate at different levels and solve different problems: cache memory hides the DRAM latency gap from the CPU, while virtual memory with paging manages the mapping between a process's logical address space and limited physical DRAM (extending it virtually using disk/SSD as backing store).

2. Comparative Table

Parameter	Cache Memory Mechanism	Virtual Memory (Paging) Mechanism
Purpose	Hides latency gap between CPU and main memory	Extends usable address space beyond physical RAM; provides isolation/protection
Managed By	Hardware (cache controller)	Hardware (MMU/TLB) + Operating System
Unit of Transfer	Cache line (typically 64 bytes)	Page (typically 4 KB, or larger with huge pages)
Hit Latency	Nanoseconds (L1: ~1 ns, L3: ~10–20 ns)	TLB hit: ~1 cycle; page table walk on TLB miss: tens of cycles
Miss Penalty	Access next cache level or DRAM (tens to ~100 ns)	Page fault requiring disk/SSD access: milliseconds (SSD) or ~10 ms (HDD) — orders of magnitude larger
Throughput Impact	High; misses are relatively cheap and frequent	Low miss tolerance; each page fault can stall a process for milliseconds, drastically reducing throughput if fault rate is high
Replacement Policy	LRU/Pseudo-LRU implemented in hardware	LRU-approximation (e.g., clock algorithm) implemented in OS software

3. Analytical Discussion

Although both mechanisms rely on the principle of locality and use a hit/miss model with replacement policies, the magnitude of their penalties differs by several orders of magnitude. A cache miss costs on the order of tens to a hundred nanoseconds because it simply escalates to the next level of on-chip or DRAM memory, whereas a page fault in virtual memory can cost milliseconds because it requires an I/O operation to secondary storage — a difference of roughly 10^5 to 10^6 in latency. Consequently, cache design tolerates relatively frequent misses, while virtual memory systems are engineered to keep page-fault rates extremely low (using large physical memory, working-set models, and prefetching).

The Translation Lookaside Buffer (TLB) acts as a small, fast cache for page table entries and is itself organized like a cache memory, illustrating how the two mechanisms intersect: a TLB hit keeps virtual-to-physical translation nearly

free, but a TLB miss requires a page table walk, and in the worst case a full page fault, directly coupling virtual memory performance to cache-like structures.

4. Impact on System Throughput

- Cache memory throughput scales with hit rate at each level; even moderate miss rates are absorbed gracefully due to the hierarchy of L1→L2→L3→DRAM.
- Virtual memory throughput is highly sensitive to page-fault rate; excessive faulting (thrashing) can collapse system throughput because the CPU sits idle waiting for slow secondary-storage I/O.
- Both mechanisms exploit temporal and spatial locality, but cache memory operates transparently at nanosecond timescales while virtual memory operates at a coarser granularity with OS involvement, trading some latency for flexibility, protection, and the ability to run programs larger than physical RAM.

5. Conclusion

Cache memory and virtual memory paging are complementary rather than competing mechanisms: cache memory optimizes latency for the CPU–DRAM gap, while virtual memory paging optimizes capacity and protection for the DRAM–disk gap. System throughput is maximized when cache hit rates are kept high (minimizing nanosecond-scale stalls) and page-fault rates are kept low (avoiding costly millisecond-scale stalls).

Q4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

1. Overview / Scope

NAND Flash and DRAM sit at different levels of the memory-storage hierarchy: DRAM serves as volatile main memory, while NAND Flash serves as non-volatile secondary storage (SSD) or, increasingly, as a storage-class memory tier. Comparing them highlights why each is suited to a different role in a hierarchical memory system.

2. Comparative Table

Parameter	NAND Flash	DRAM
Volatility	Non-volatile (retains data without power)	Volatile (loses data without power)
Cell Structure	Floating-gate/charge-trap transistor storing charge as bits	1T-1C capacitor-based cell
Read Latency	~25–100 μs (microseconds)	~50–100 ns (nanoseconds) — roughly 1000× faster than NAND
Write/Program Latency	~200–900 μs, plus block erase-before-write overhead	~50 ns, no erase requirement
Bandwidth	Sequential: ~500 MB/s–7 GB/s (NVMe SSD); random access much lower	Tens of GB/s per channel (DDR5 ~40–50 GB/s); high for both sequential and random
Endurance	Limited program/erase (P/E) cycles (~1,000–100,000 depending on type)	Effectively unlimited write endurance
Granularity	Read/write in pages (~4–16 KB); erase in blocks (~MB)	Read/write at byte/word granularity
Cost per Bit	Low (cheaper than DRAM, more than HDD)	Higher than NAND Flash
Role in Hierarchy	Secondary storage / storage-class memory tier	Primary (main) memory

3. Analytical Discussion

The fundamental difference driving these gaps is physical: DRAM cells are accessed electrically at nanosecond speed with no wear mechanism, while NAND Flash stores charge in a floating gate that must be read by sensing threshold voltage and written by tunneling electrons through an oxide layer — a slower and wear-inducing process. This is why NAND Flash requires block erase before rewriting a page (unlike DRAM's direct overwrite) and why NAND has finite program/erase endurance, whereas DRAM does not degrade with repeated writes.

Bandwidth-wise, modern NVMe SSDs achieve impressive sequential bandwidth through massive internal parallelism (multiple channels and dies), but DRAM still leads decisively in random-access bandwidth and, more importantly, in latency consistency, because DRAM has no controller-level translation layer (FTL) or garbage collection overhead that can introduce unpredictable latency spikes as NAND Flash does.

4. Suitability for Hierarchical Memory Systems

- DRAM is suited as main memory because its nanosecond latency and byte-addressability match the CPU's need for fast, fine-grained, unlimited-endurance access to active data.
- NAND Flash is suited as persistent secondary storage because its non-volatility retains data without power, its cost per bit is far lower than DRAM, and its block-oriented access pattern fits large, less latency-sensitive file and page-swap operations.
- In emerging hierarchies, NAND Flash (or Storage-Class Memory /NVMe tiers) is increasingly used as a caching or swap tier below DRAM, extending effective capacity for memory-intensive workloads at a latency cost far lower than traditional HDDs but still far higher than DRAM.

5. Conclusion

NAND Flash and DRAM are not interchangeable but complementary: DRAM's speed and byte-level access make it ideal for the CPU-facing working memory tier, while NAND Flash's non-volatility, density, and lower cost make it ideal for persistent, larger-capacity storage lower in the hierarchy. Their combination underlies virtual memory paging and modern tiered-storage architectures.

Q5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

1. Overview / Scope

MRAM (Magnetoresistive RAM) and RRAM (Resistive RAM, also called ReRAM) are emerging non-volatile memory (NVM) technologies positioned as candidates to bridge the gap between fast-but-volatile SRAM/DRAM and dense-but-slow NAND Flash. Both aim to combine non-volatility with near-DRAM speed, but they differ in their underlying physical storage mechanism and resulting performance profile.

2. Comparative Table

Parameter	MRAM (Magnetoresistive RAM)	RRAM (Resistive RAM)
Storage Mechanism	Magnetic tunnel junction (MTJ); data stored as electron spin/magnetic orientation	Resistance switching in a metal-oxide layer via formation/rupture of conductive filaments
Access Time	Very fast, ~2–20 ns (STT-MRAM), close to DRAM/SRAM	Moderate-fast, ~10–50 ns, generally slower than MRAM
Write Endurance	Very high (~10 ¹² –10 ¹⁵ cycles), near-DRAM class	Moderate (~10 ⁶ –10 ¹² cycles), lower than MRAM
Throughput	High, suitable for cache/DRAM-replacement roles	Good, but somewhat lower sustained throughput due to slower switching/settling time
Energy Efficiency	Low read energy; write energy moderate (current-driven switching)	Very low read/write energy per bit; but variability can add error-correction overhead
Non-volatility	Yes (retains data without power)	Yes (retains data without power)
Scalability/Density	Moderate; MTJ stack limits extreme scaling	High; simple two-terminal structure scales well, supports 3D crossbar stacking
Maturity/Use Case	Used in embedded NVM, some cache/last-level applications	Promising for storage-class memory and neuromorphic/in-memory computing

3. Analytical Discussion

MRAM stores information using the relative magnetic orientation of two ferromagnetic layers separated by a thin insulator (a magnetic tunnel junction); switching this orientation via spin-transfer torque (STT) allows very fast, low-latency writes with extremely high endurance, since no physical material is being altered — only its magnetic state — which is why MRAM endurance approaches that of DRAM. RRAM instead relies on the physical formation and rupture of a conductive filament within a metal-oxide layer to switch between high- and low-resistance states; this physical filament change is inherently more variable and wears the material over repeated cycles, giving RRAM lower endurance and somewhat higher latency variability than MRAM, but its simple two-terminal structure makes it easier to scale to very high densities and stack in 3D crossbar arrays.

In terms of energy efficiency, both technologies benefit from zero standby (leakage) power since they are non-volatile, unlike SRAM and DRAM which consume power continuously to retain data or perform refresh; however, RRAM generally achieves lower per-bit switching energy, making it attractive for extremely dense, low-power applications, while MRAM's advantage lies in its very fast, reliable, high-endurance switching, which suits it to more frequently-written, performance-critical roles.

4. Comparative Performance Summary

- Access time: MRAM is generally faster and more consistent (closer to DRAM), while RRAM is fast but can show greater cycle-to-cycle variability.
- Throughput: MRAM is better suited to high-frequency, cache-like or DRAM-adjacent roles; RRAM is better suited to high-density storage-class memory where extreme speed is secondary to density.
- Energy efficiency: RRAM typically offers lower per-bit energy for read/write due to its simple structure, while MRAM offers a strong balance of speed and endurance with moderate energy use, avoiding refresh power entirely (unlike DRAM).
- Endurance and reliability: MRAM's endurance (near-DRAM class) outclasses RRAM's, making MRAM more suitable where data changes very frequently, whereas RRAM's density advantage suits it for bulk, less frequently rewritten storage.

5. Conclusion

MRAM and RRAM both represent promising non-volatile alternatives capable of narrowing the latency-versus-persistence trade-off in hierarchical memory systems. MRAM leans toward speed and endurance (a DRAM/cache-class role), while RRAM leans toward density and energy efficiency (a storage-class memory role). Their continued development is expected to reshape memory hierarchies by inserting a persistent, near-DRAM-speed tier between volatile RAM and traditional Flash-based storage.